

How to Start DSA (Data Structures & Algorithms) – Practice-Based Guide

1ST OF ALL THERE IS NO SHORTCUT TO IT (OR ANYTHING). IF YOU WANT TO DO SOMETHING YOU HAVE TO WORK FOR IT

1. Pick a Structured Roadmap

- Use the **NeetCode 150 sheet** (or similar curated lists) as your main reference.
 - This ensures you cover all important DSA topics in a structured order instead of solving problems randomly. (**NEVER SOLVE RANDOMLY**)
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2. Topic-by-Topic Approach

For each topic (e.g., Arrays → Two Pointers → Sliding Window → Stack → Queue → ...):

1. Learn the concept briefly (from YouTube, NeetCode video, or your course notes). Do not over-theorize—just get the basics.
 2. Open LeetCode and filter problems from that topic.
 3. Progress from easy → medium. (don't start hard until you are done with DSA or are fully confident in your practice)
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3. How to Practice Problems

- Timebox yourself:
 - Spend at least **2 hours** on a problem before peeking at solutions.
 - Try all possible approaches you can think of (even brute force), even guess, heck most problems are based on such guesses.
 - When stuck:
 - If after 2 hours you cannot progress, then look at hints or editorial.
 - **Do not copy-paste solutions—re-implement in your own words.**
 - Repeat until you fully understand the solution.
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4. Active Learning Techniques

- Maintain a DSA notebook (digital or physical):
 - Note down patterns (e.g., "sliding window often solves substring problems").
 - Record brute force → optimized approach → final solution.
 - Teach back: explain your solution out loud or to a peer.
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5. Patterns Over Problems

- Instead of memorizing individual problems, learn **problem-solving patterns** (two-pointer, binary search, dynamic programming states, graph traversal, etc.).
 - Each LeetCode problem teaches you a pattern that reappears in harder problems.
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6. Weekly Schedule (Sample)

- Monday to Friday: 2–3 problems per day (focused on current topic).
 - Saturday: Revisit problems you failed earlier in the week.
 - Sunday: Do a mixed-topic mock session (pick 3–5 problems randomly).
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7. Consistency and Tracking

- Use a tracker (Notion, Excel, or Google Sheets) to mark:
 - Solved by myself
 - Needed hint
 - Could not solve
 - Revisit “needed hint” and “could not solve” problems after 1–2 weeks until you can solve them unaided.
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8. When to Move On

- Do not rush: solve 5–10 solid problems per subtopic before moving to the next.
 - You do not need to solve every LeetCode problem—focus on representative ones.
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9. Mindset

- $\text{DSA mastery} = \text{time} \times \text{consistency} \times \text{deliberate practice}$.
- Struggling for hours is normal—your brain is learning patterns.
- The goal is to think like a problem solver, not just grind problems.

References to Learn DSA

All resources are good. Some are liked by one group, others by another. The most loved ones are listed here:

Video Lectures (Concept Building)

- **DSA by Abdul Bari (YouTube)** – most loved for crystal-clear basics.
- **NeetCode (YouTube)** – short and to-the-point explanations with direct mapping to LeetCode problems.
- **Take U Forward (YouTube by Striver)** – excellent for structured playlists and explanations in C++/Java/Python.
- **MIT OpenCourseWare – Introduction to Algorithms** – for those who want deep theoretical understanding.

Practice Platforms

- **LeetCode** – main platform for topic-based and interview-style problems.
- **Codeforces / AtCoder / CodeChef** – great for building speed and problem-solving for contests.

Roadmaps and Sheets

- **NeetCode 150 / 75** – most popular structured DSA roadmap for interviews.
- **Striver's SDE Sheet** – covers DSA, system design, and interview prep.
- **Blind 75** – minimal must-do problems for interviews.

Extra for Specific Learning Styles

- If you prefer visuals: **VisuAlgo.net** – interactive animations for data structures and algorithms.

- If you prefer reading: CP-Algorithms – competitive programming–style explanations.
 - If you prefer practicing patterns: AlgoMonster – focuses on patterns behind problems.
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How to Use References Efficiently

1. Learn concept → watch video or read article
2. Open LeetCode sheet → practice 5–10 problems from that topic
3. If stuck → peek at editorial or NeetCode video → re-implement
4. Document → write summary and solution pattern in your notebook

TIPS

1. Try to do the problem of the day on leetcode.
2. Do try out leetcode contests and atcoder ABC contests.
3. **Do not overdo leetcode:** just complete all topics and neetcode sheets and shift to atcoder and codeforces.
4. If you want to improve more than typical DSA, focus on atcoder and codeforces you can also give leetcode contests.